



UNIVERSITY OF
ILLINOIS
URBANA-CHAMPAIGN

SE 423: Introduction to Mechatronics

Lecture 22: 3D Vision, Structure from Motion, and Deep Depth

Marius Juston

Monday, April 13th, 2026

Talk to the person next to you and answer these questions.

- Why do we use CIE Lab color space instead of RGB for object tracking?
- How do we calculate the centroid of a binary blob using Image Moments?
- If a red ball enters a dark shadow, which CIE Lab parameter drops? Which ones remain constant to help us track it? (You can look at the website https://marius-juston.github.io/SE-423---Class-Material/color_spaces.html) for a reminder.
- Put an object in front of yourself, grab it with both eyes open, now close one eye and try to grab it again. Did you feel like it was a different distance than expected?

Office Hours:

- **Monday:** 4-6 PM, Neel
- **Tuesday:** 11-1 PM, Lakshmi
- **Tuesday:** 1-3 PM, Samuel
- **Tuesday:** 2-5 PM, Dan & Marius

Lab: Starting Lab 7 this week. This a 2-week lab! This is the final lab! Afterwards you will just be working on your final project in your teams.

Make sure that you fill in the Google Sheet from the Announcement:

<https://docs.google.com/spreadsheets/d/1lh2lSsxLq3i58cWuxZ1K7XvA4rPz2CyiUasCmF61Gnc/edit?usp=sharing>

Homeworks:

- Remainder of [homework 5](#) due on Gradescope on **April 21, 5PM**
- Next homework is primarily the **personal project** due on **April 22, 9AM**.
 - I would highly recommend you thinking now and maybe even starting it now if you have a good idea already!

In the class material questionnaire, a lot of you did not specify if there was a specific application of AI / Computer Vision / Algorithms that you were interested in.

For the people that come to class what would you find useful for the final project?

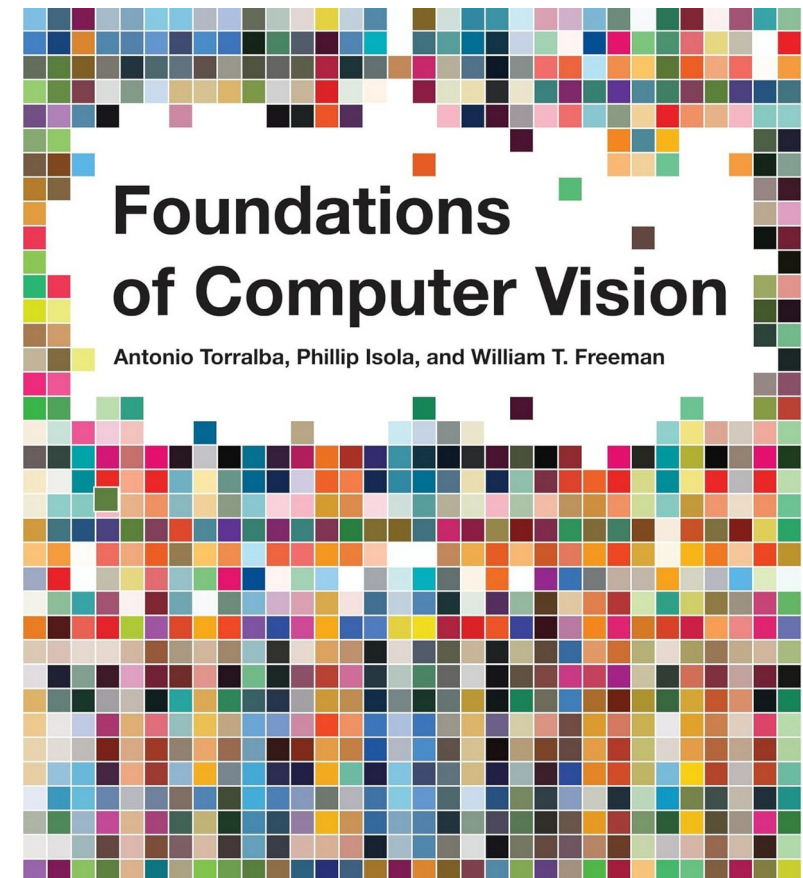
Questions?

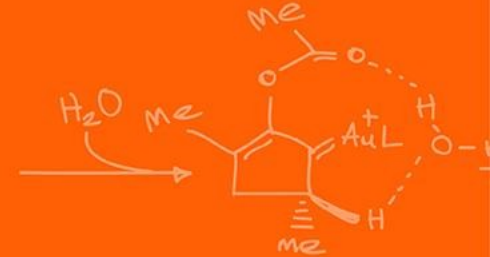
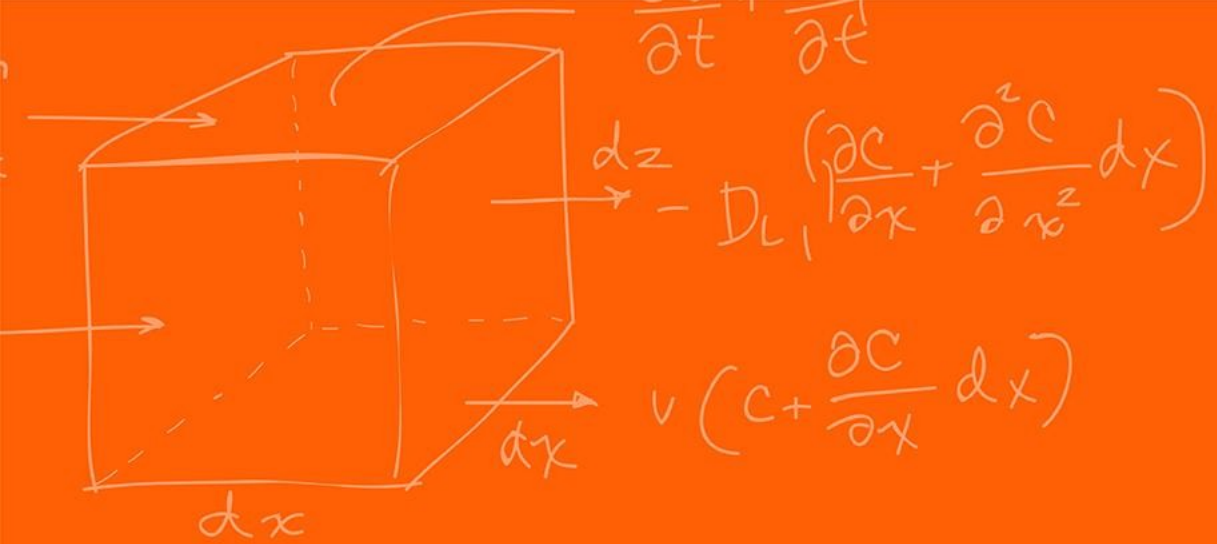
Homework, Lab, LabVIEW?

Again I would **HIGHLY** recommend you looking at:

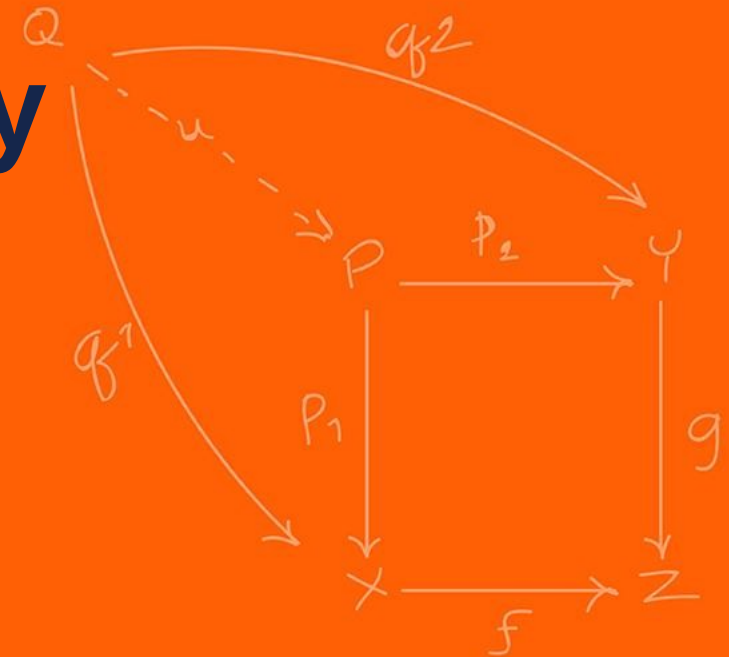
- <https://visionbook.mit.edu/>
- The webpage is a textbook with all the math, background and information for computer vision.
- We will only be doing a tiny portion of this in this lecture (there are whole grad classes for computer vision).
- It is containerized very nicely, has lots of pictures, diagrams. Each section takes 5 minutes to read.

All the material and pictures were taken there today





Camera Geometry



Cameras map the 3D world into a 2D image plane.

Is this a lossy or lossless transformation? If lossy give an example.

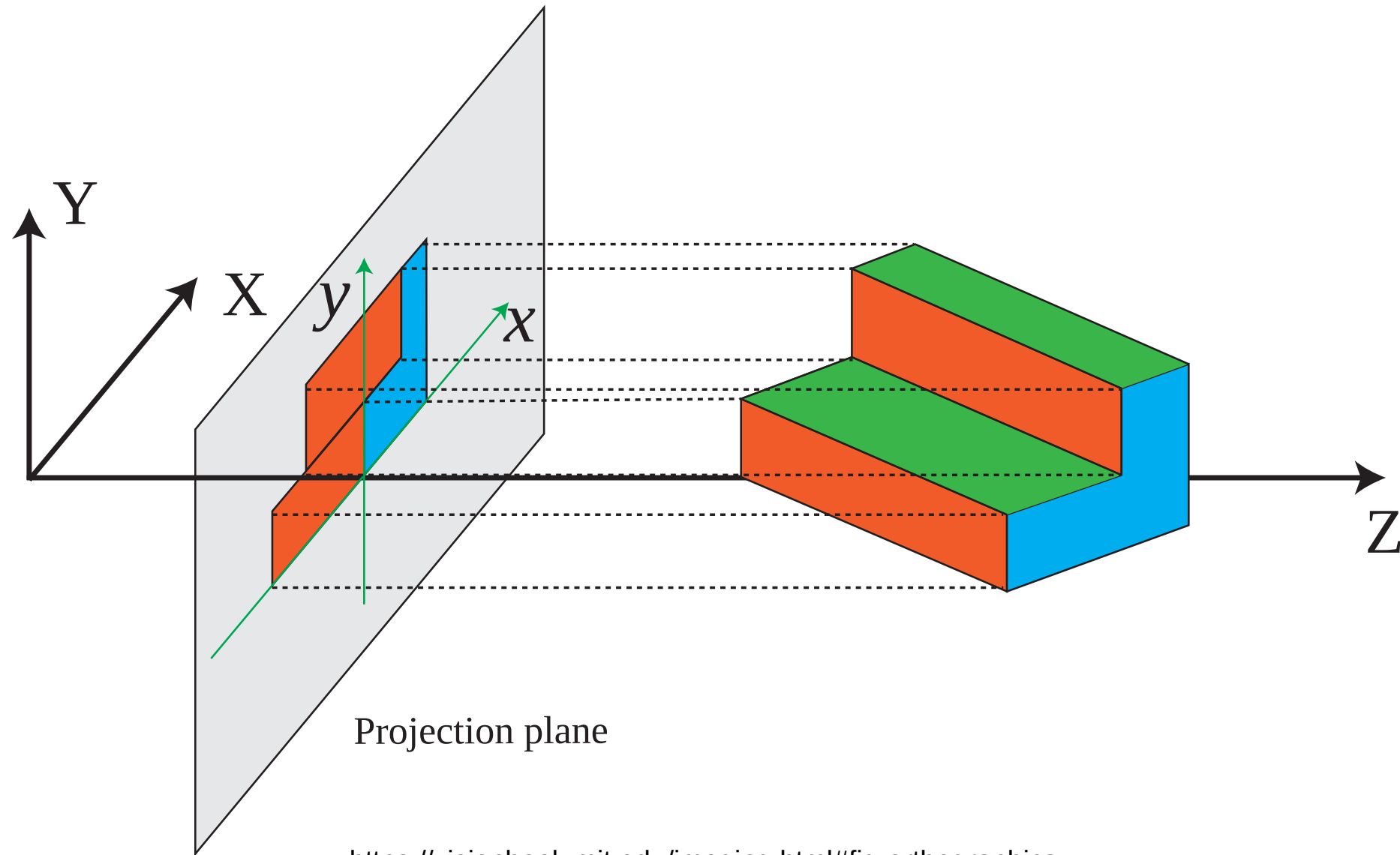
Infinite points on a single ray maps to the exact same pixel.

However, we would like to control robots in 3D space, as such we must invert the projection.

The first step is representing the model of the camera.

If infinite 3D points map to one pixel, is the inverse function mathematically solvable with a single image?

The 3D to 2D project



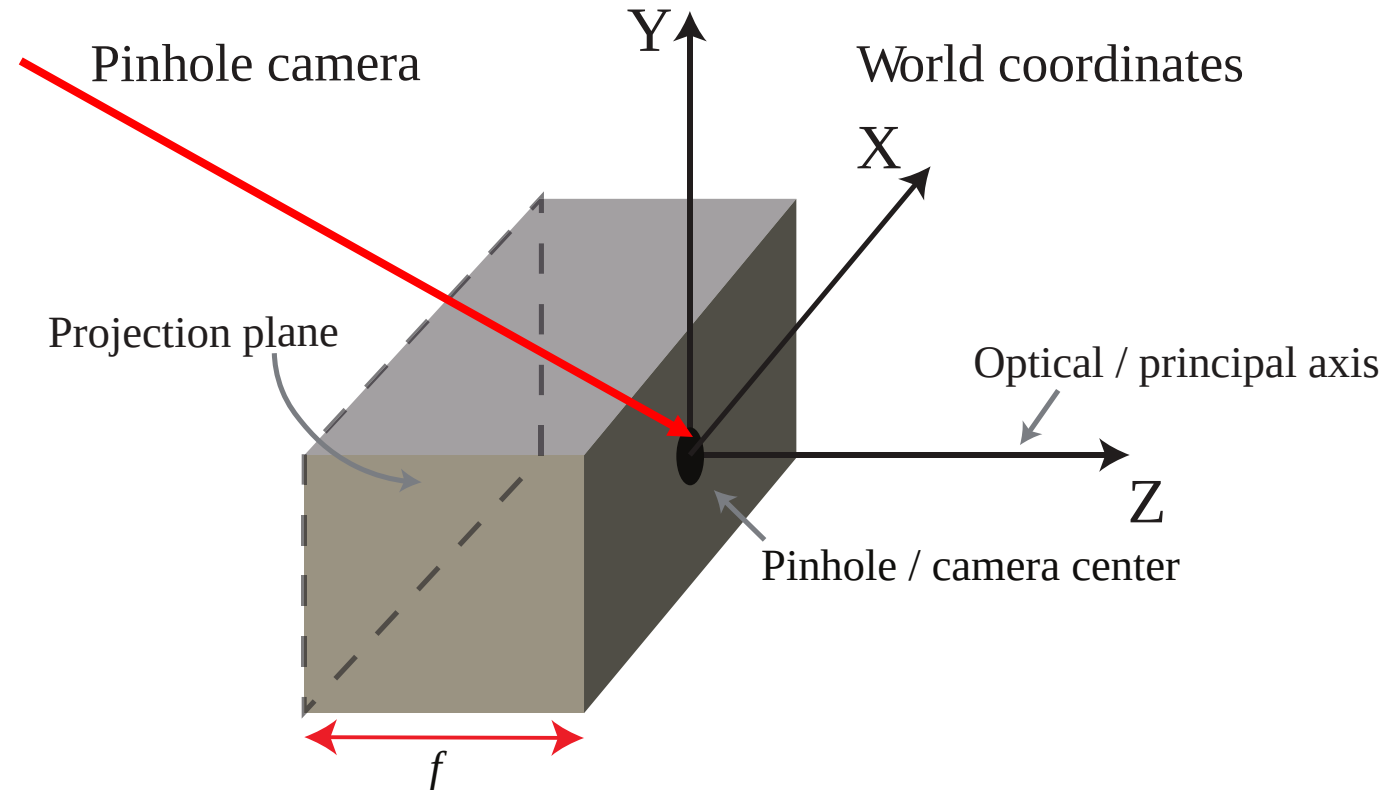
<https://visionbook.mit.edu/imaging.html#fig-orthographics>

A pinhole camera projects 3D coordinates in the world to 2D positions on the projection plane of the camera through the straight-line path of each light ray through the pinhole.

There are 3 coordinate systems that we will use.

World coordinates: Let the origin of a World Cartesian coordinate system be the camera's pinhole.

The coordinates of 3D position of a point, P , in the world will be $P = (X, Y, Z)$, where the Z axis is perpendicular to the camera's sensing plane (projection plane).

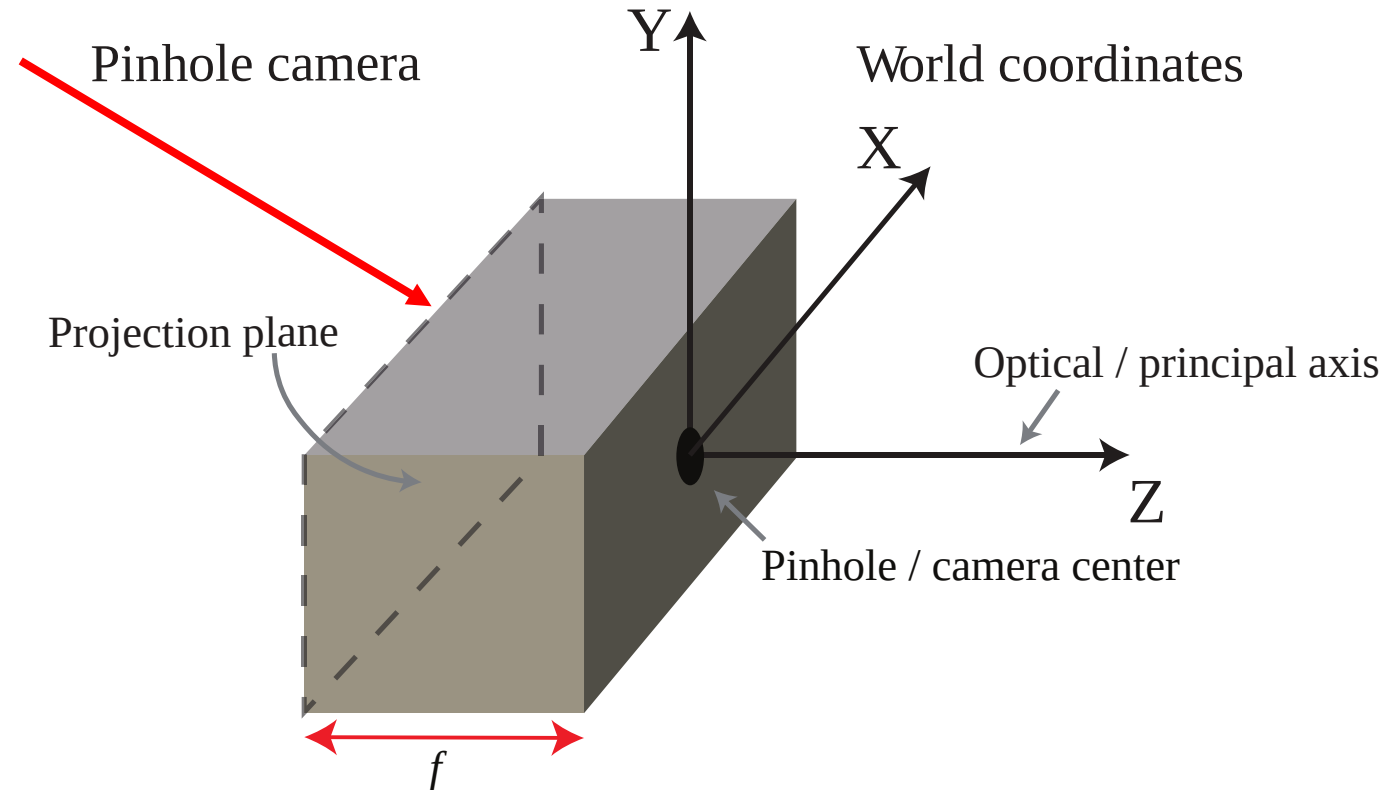


<https://visionbook.mit.edu/imaging.html#fig-orthographics>

Camera coordinates in the virtual camera plane.

The camera projection plane is behind the pinhole and at a distance f of the origin.

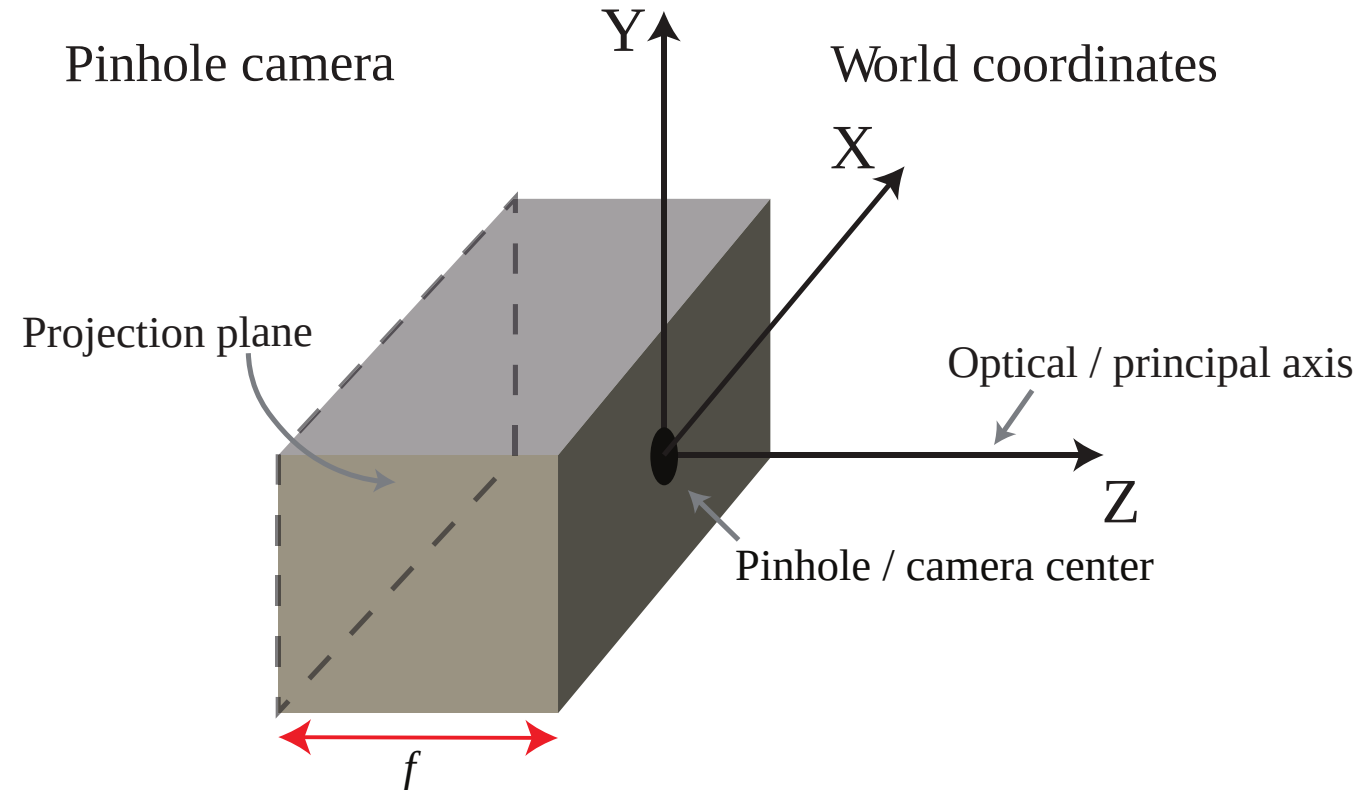
Because the rays get inverted through the pinhole, it is useful to reflect the project plane so that the coordinates are easier to understand



<https://visionbook.mit.edu/imaging.html#fig-orthographics>

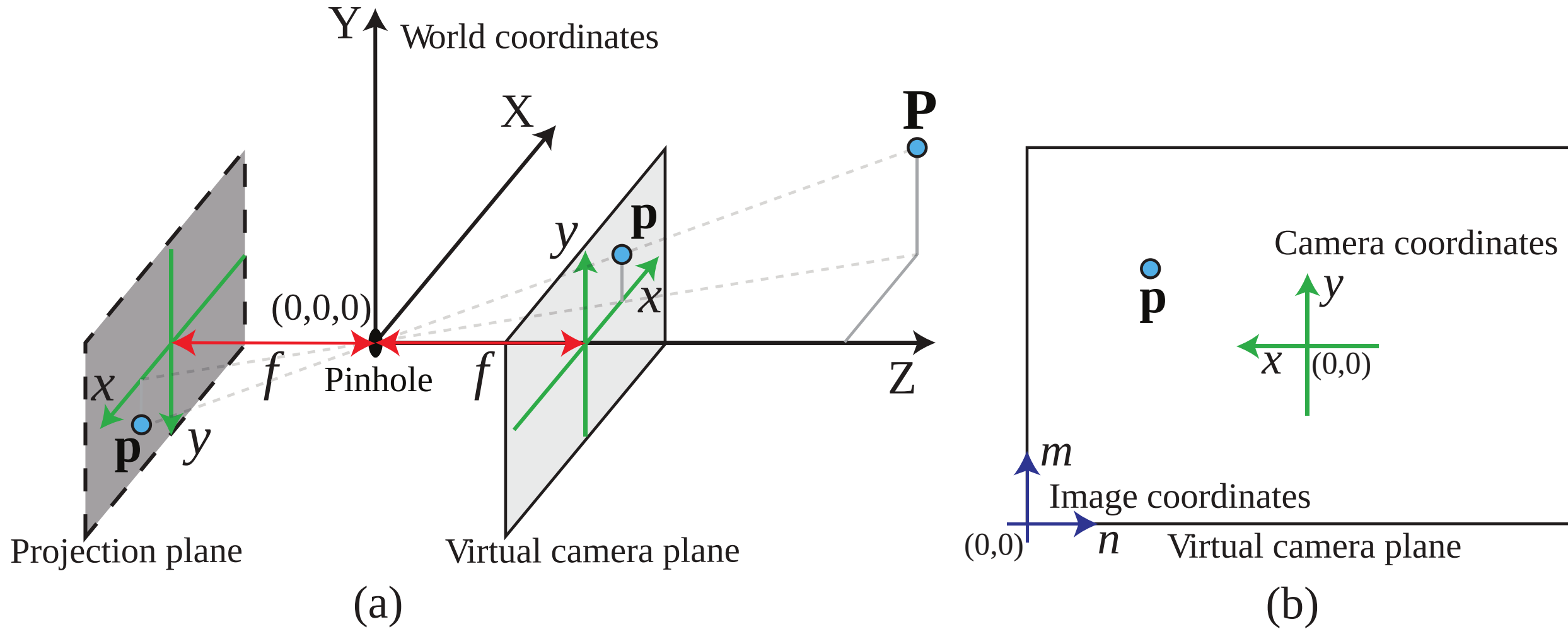
Image coordinates, are typically measured in pixels.

Both the camera coordinate system (x, y) and the image coordinate system (n, m) are related by an affine transform as we will discuss later.



<https://visionbook.mit.edu/imaging.html#fig-orthographics>

The 3D to 2D project



<https://visionbook.mit.edu/imaging.html#fig-orthographics>

The pinhole model assumes the light passes through a single infinitively small aperture.

By similar triangles, a 3D point (X, Y, Z) , projects to the image plane at (x, y) .

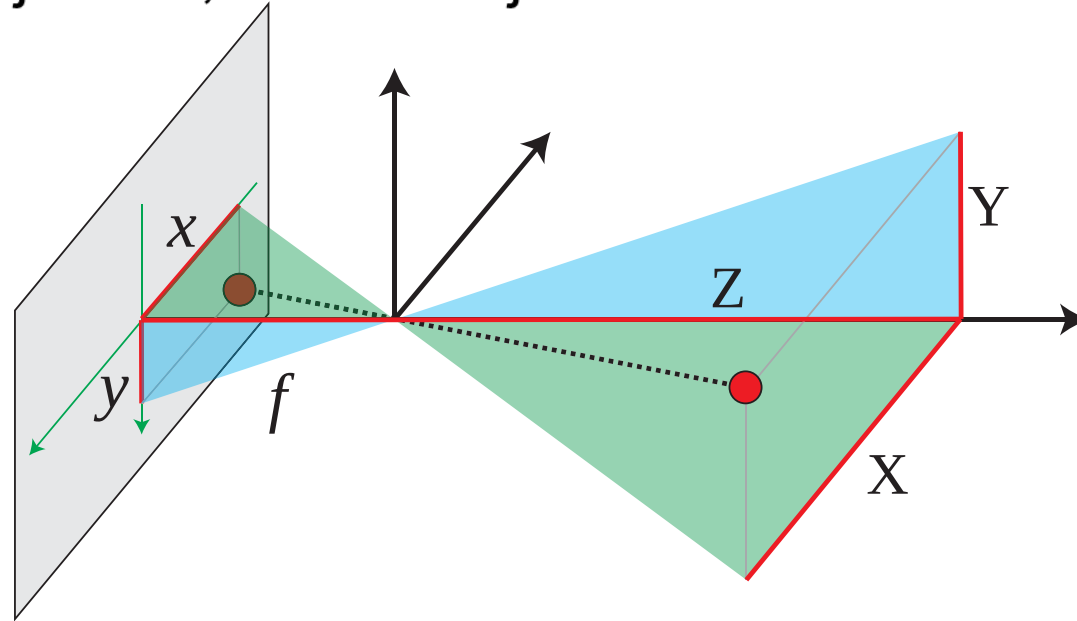
The ideal projection equations are:

$$x = f \frac{X}{Z}, \quad y = f \frac{Y}{Z}$$

Where f , represents the focal length.

These equations are called the **perspective projection equations**.

Under perspective projection, distant objects become smaller, through the inverse scaling by Z .



The perspective projection equations apply not just to pinhole cameras but to most lens-based cameras, and human vision as well. The pixel coordinates can be computed as,

$$n = -ax + n_0, \quad m = ay + m_0$$

Homogeneous coordinates represent a vector of length N by a vector of length $N + 1$. The transformation rule from **heterogeneous** to **homogeneous** coordinates is simply to add an additional coordinate, 1, to the heterogeneous coordinates as shown here:

$$\begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

To go from homogeneous coordinates back to the **heterogeneous** representation of the point, we divide all the entries of the homogeneous coordinate vector by their last dimension

$$\begin{bmatrix} x \\ y \\ w \end{bmatrix} \rightarrow \begin{pmatrix} x/w \\ y/w \end{pmatrix}$$

As we have already discussed, the perspective projection equations are nonlinear and involve a division by the point's Z -location.

That is, a 3D point with coordinates (X, Y) appears in the image at coordinates

$$x = f \frac{X}{Z}, \quad y = f \frac{Y}{Z}$$

This division complicates algebraic relations.

Projective geometry and homogeneous coordinates help that simplify those algebraic relations and thus simplify many of the equations that involve geometric transformations.

Consider a coordinate system with the origin fixed at the center of projection of the camera and the 3D point in homogeneous coordinates,

$$P = \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

We can verify that the that matrix, K , multiplies, by the vector describing the 3D P , returns the position of its **perspective projection**!

We thus can convert the 3D coordinate P , to it's projected 2D position p !

$$KP = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \begin{bmatrix} fX \\ fY \\ Z \end{bmatrix} = \lambda \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Transforming the product vector from homogeneous coordinates to heterogeneous coordinates yields

$$\begin{bmatrix} fX \\ fY \\ Z \end{bmatrix} \rightarrow \begin{pmatrix} fX/Z \\ fY/Z \end{pmatrix} = \begin{pmatrix} cx \\ y \end{pmatrix} = p$$

Camera Intrinsics

This formulation is one of the biggest benefits of using homogeneous coordinates. It allows writing perspective projection as a **linear operator** in homogeneous coordinates.

We have the world in pixels but that is not very useful! We want to use an actual measurement system such as metric or feet.

To do this we need to find the **pixel to measurement unit** relationship!

Cameras translate the world into pixels, but the relationship between the 3D scene and the 2D image captured by the camera is not always straightforward.

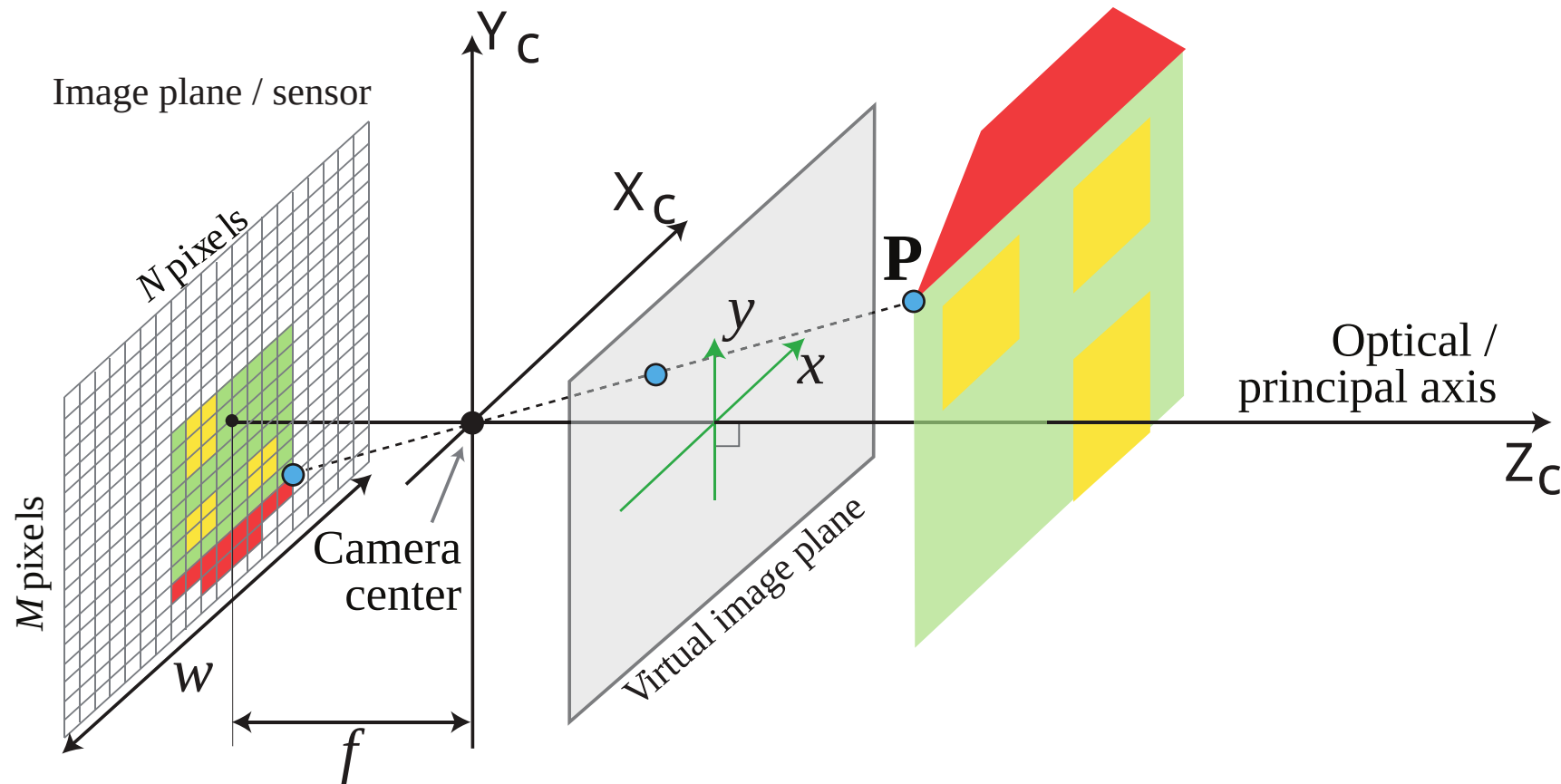
To start with, the world is measured in units of meters while points in images are measured in pixels.

Factors such as geometric distortions can influence this relationship, and we need to account for these elements in our camera model.

The Pinhole Camera Model



We need to account for the perspective projection of the point, P , described in camera coordinates, to pixel coordinates in the sensor plane of the camera.



The units of the position in pixel space may be different than those of the world coordinates, which scales the focal length, f , to some other value. Let's call that value a .

$$K = \begin{bmatrix} a & 0 & 0 & 0 \\ 0 & a & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Where the constant a , is related to the physical camera parameters:

$$a = f N / w$$

Where f is the focal length (in meters), w is the width of the sensor in meters, and N is the image width in pixels. The ratio N / w is the number of pixels per meter in the sensor

Also, the image center might not be the pixel $[0,0]^T$. If the optical axis coincides with the image center, we need to apply a translation so that a 3D point on the optical axis (the location of the pinhole, $X = Y = 0$) projects to $[c_x, c_y]^T$.

For example, for an image of size $N \times M$, we will have $c_x = N/2$ and $c_y = M/2$

$$K = \begin{bmatrix} a & 0 & c_x & 0 \\ 0 & a & c_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

It is also possible that the pixels are not square, in which case we must account for different scaling along both axis:

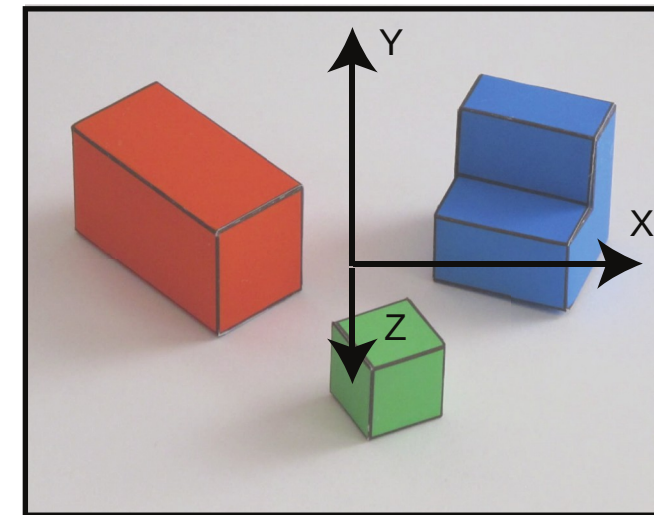
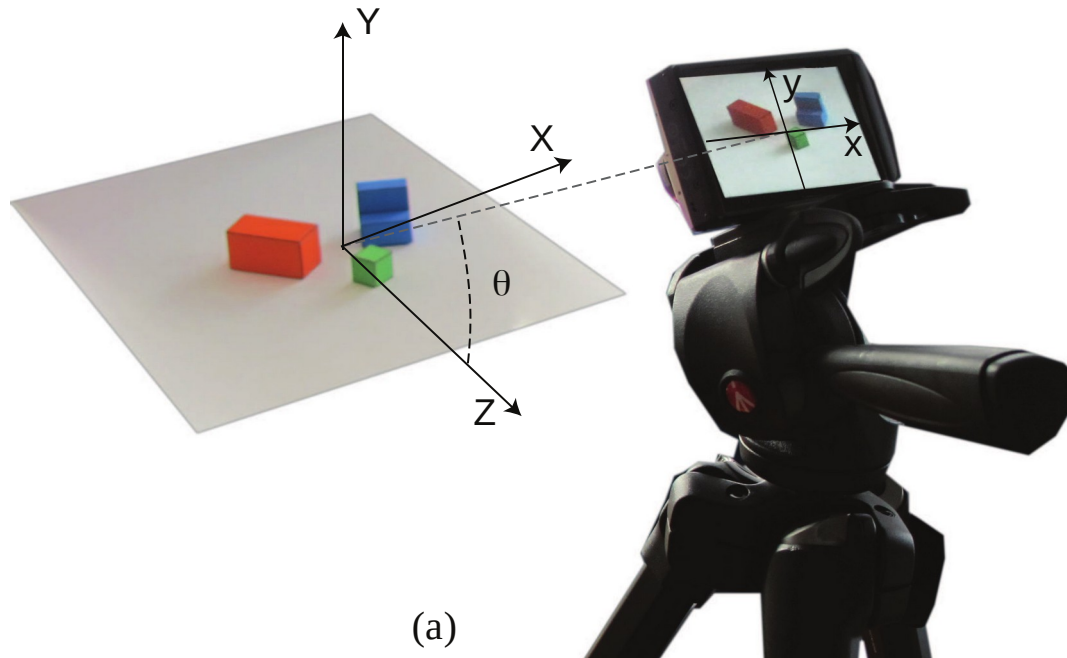
$$K = \begin{bmatrix} a & 0 & c_x & 0 \\ 0 & b & c_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Why might a and b , be slightly different on cheap CMOS sensors?

Camera Extrinsics

In all our previous examples, we have been assuming that the camera origin was located at the origin of the world coordinate system with the optical axis aligned with the world Z-axis.

In practice, there will be many situations where placing the camera at a special position will be more convenient.

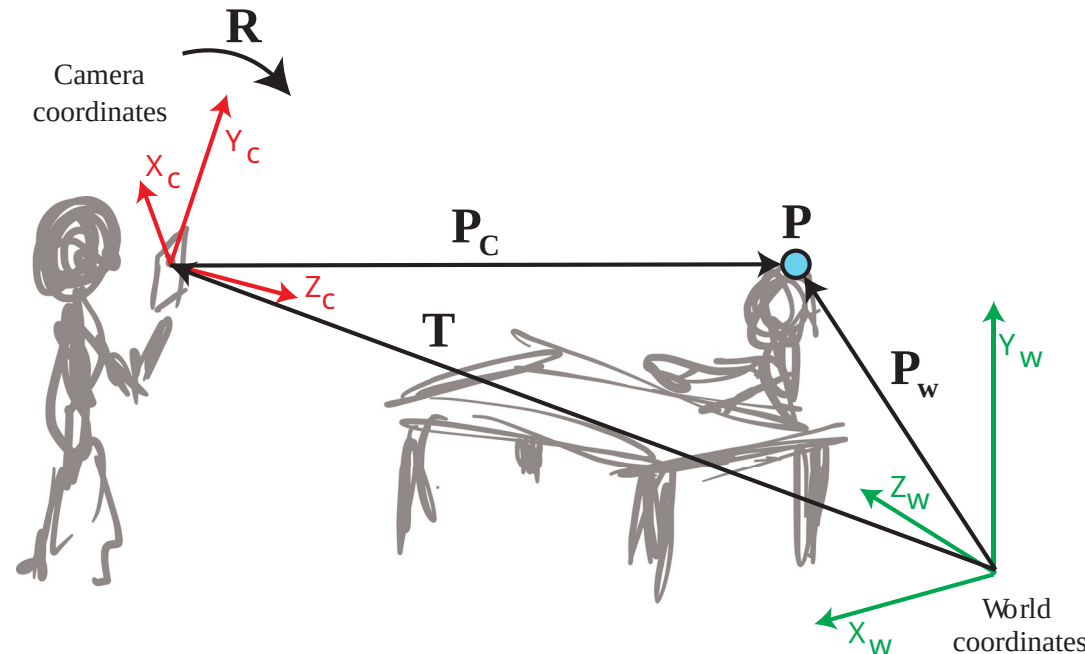


The Pinhole Camera Model



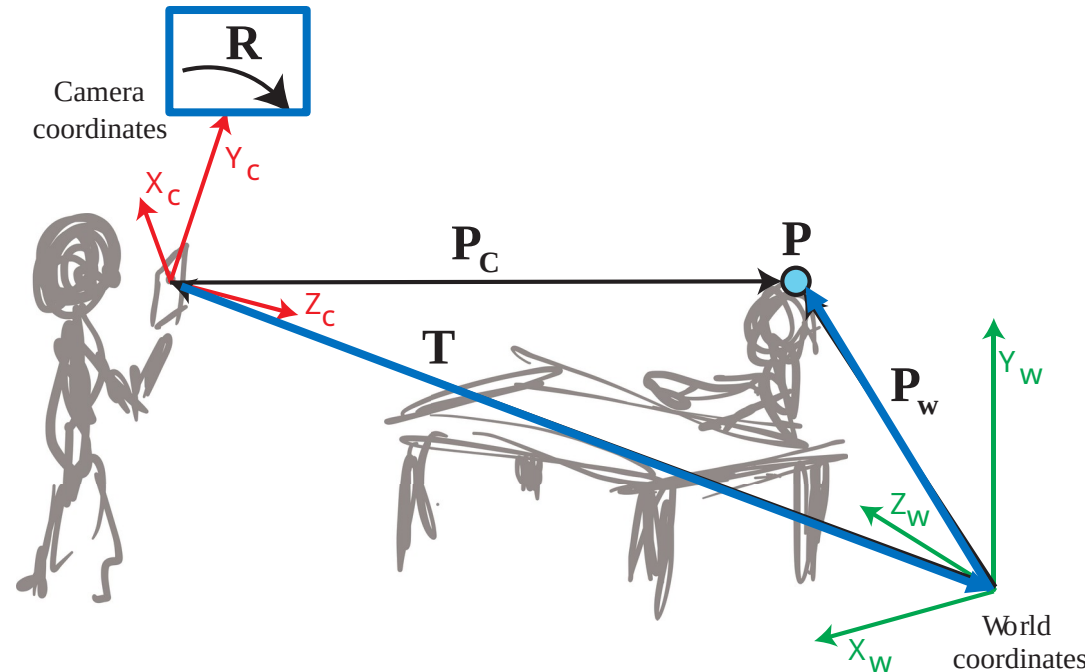
Let's say that our camera has its camera center displaced by vector T and the axes are rotated by R^T with respect to the world-coordinates system.

We are now given a point in the world P_w , with its coordinates are described in terms of the world-coordinate system. Our goal is to find how that point projects into the image captured by the camera.



Based on coordinate transformations of the camera, we can describe the transformation as,

$$P_c = R(P_w - T) = RP_w - RT$$



Using homogeneous coordinates, to describe the transformation from world coordinates to coordinates, the result is,

$$P_C = \begin{bmatrix} R & -RT \\ 0^T & 1 \end{bmatrix} P_W$$

The camera parameters, R and T, that relate the camera to the world are called the **extrinsic camera** parameters. These parameters are external to the camera.

We have seen now how to transform a point described in the **world-coordinate** system into the **camera-coordinates** system. We can now use the intrinsic camera parameters to project the **point** into the **image plane**.

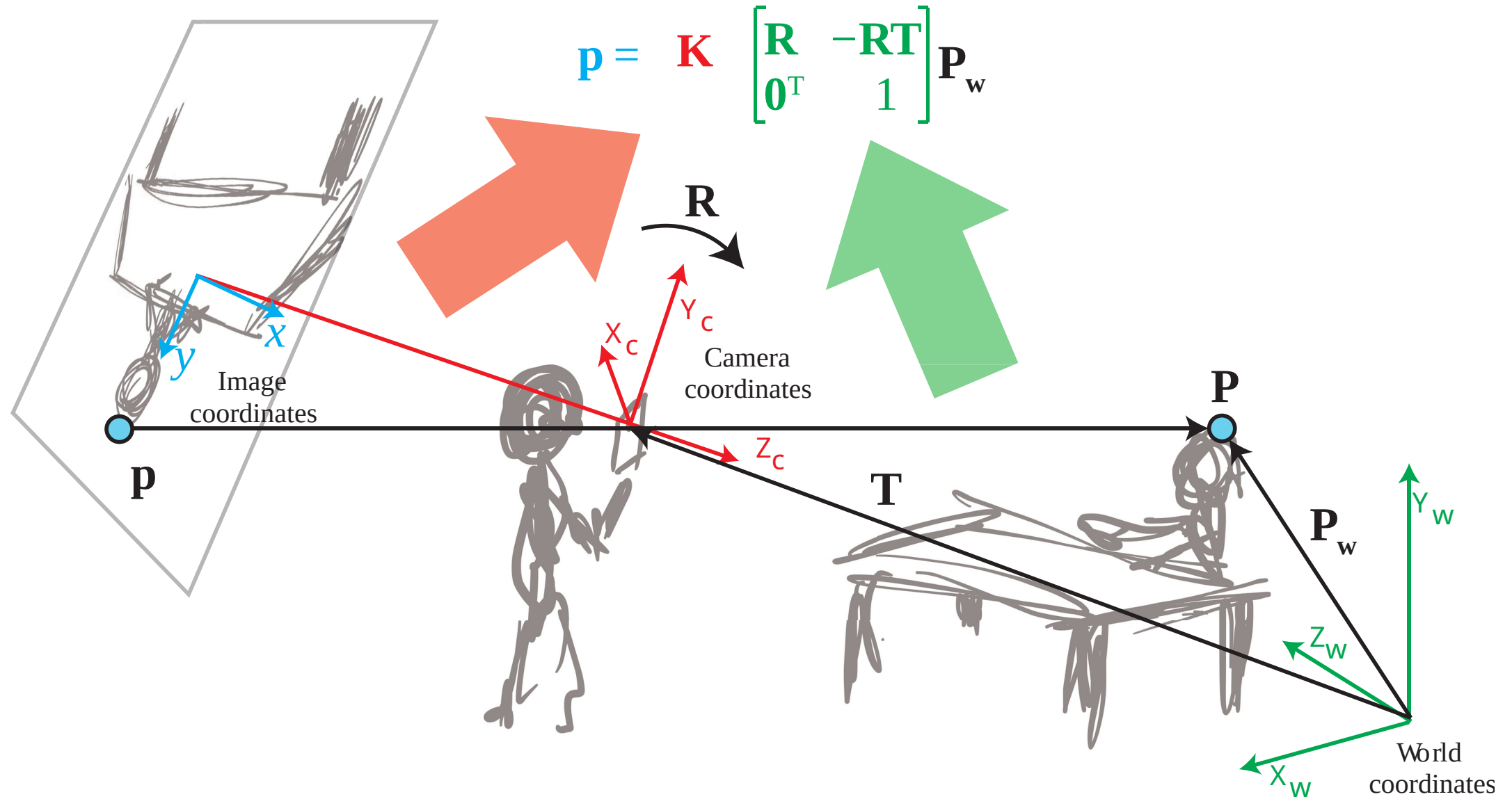
The concatenation of the **extrinsic** and **intrinsic** transformation matrices yields the desired **transformation matrix**, which will transform world coordinates to rendered pixel coordinates:

$$P_C = K \begin{bmatrix} R & -RT \\ 0^\top & 1 \end{bmatrix} P_W$$

How many degrees of freedom does the extrinsic matrix have?

The full 6! Translation + Rotation in 3D

The Pinhole Camera Model



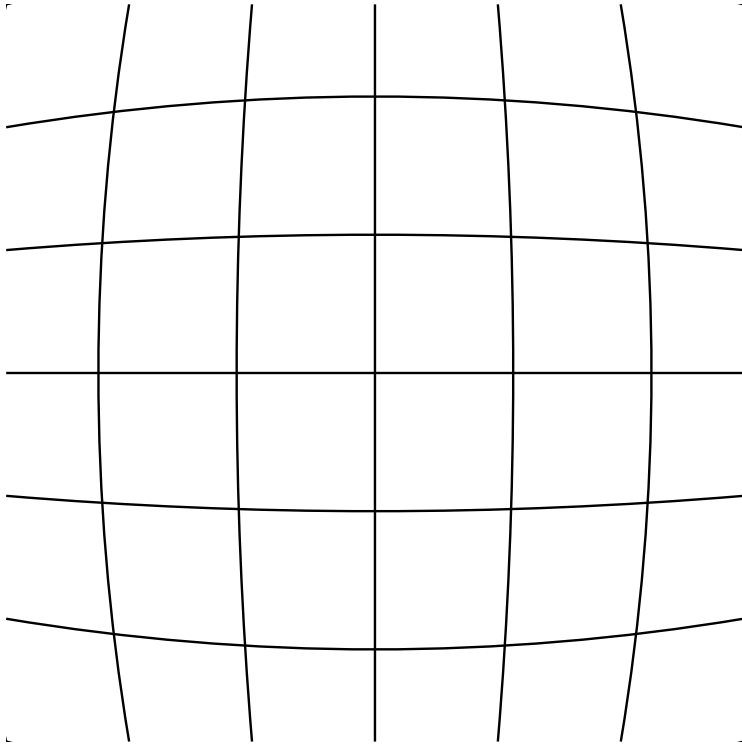
Real lenses are not perfect pinholes; they bend light, causing distortion.

Radial distortion (barrel/pincushion) occurs due to spherical lens shapes.

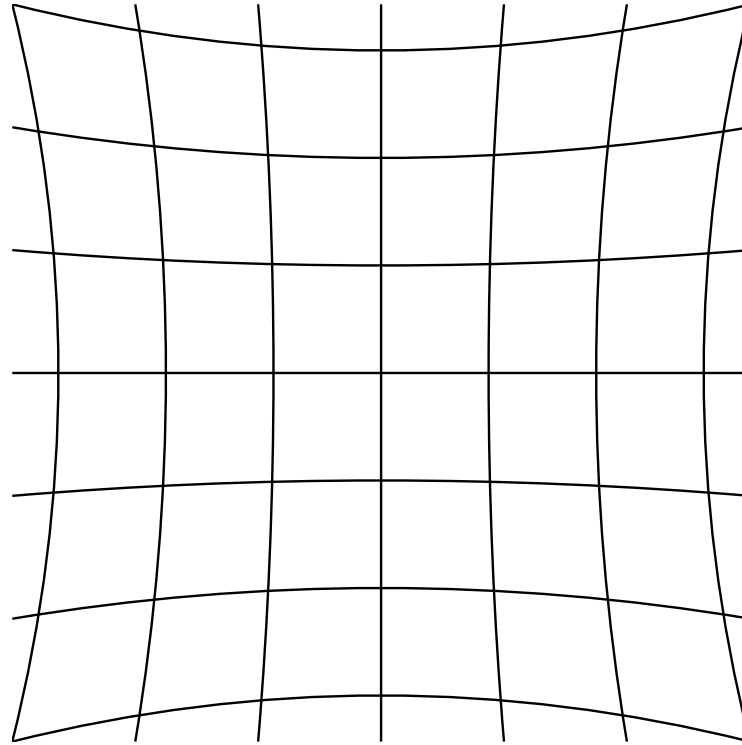
Tangential distortion occurs if the lens is not perfectly parallel to the CMOS sensor.

Why do we need to undistort an image before doing 3D geometry calculations?

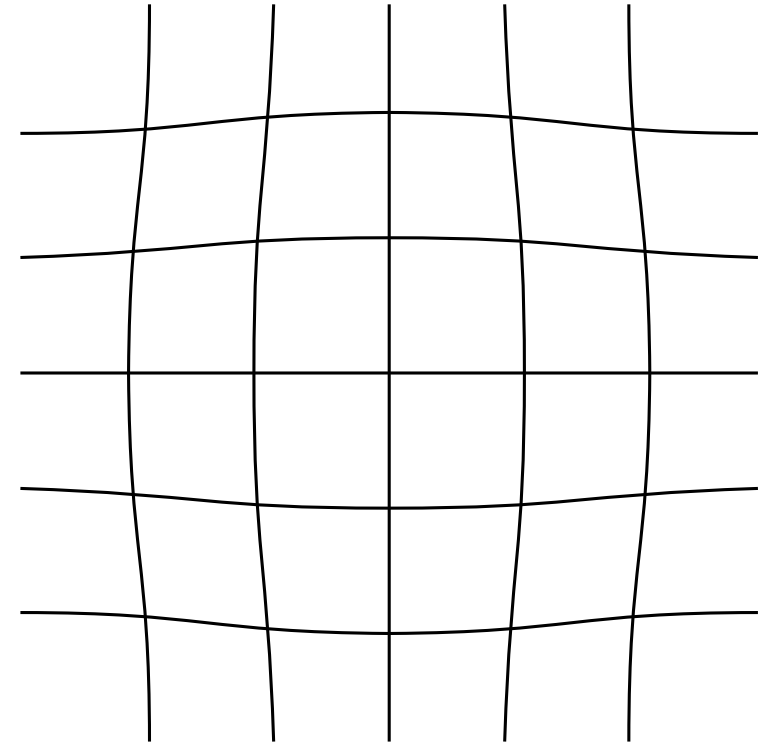
We assume that all our math is planar and linear. If we have the distortions, we have non-linear systems and thus our math no longer works!



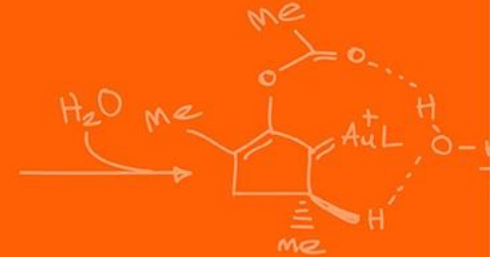
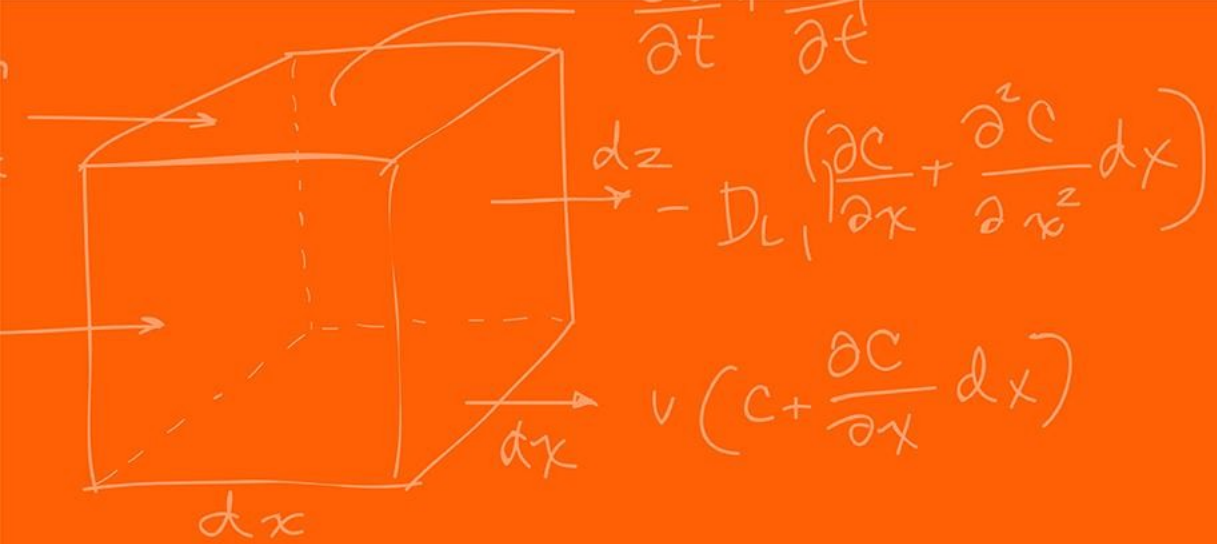
Barrel



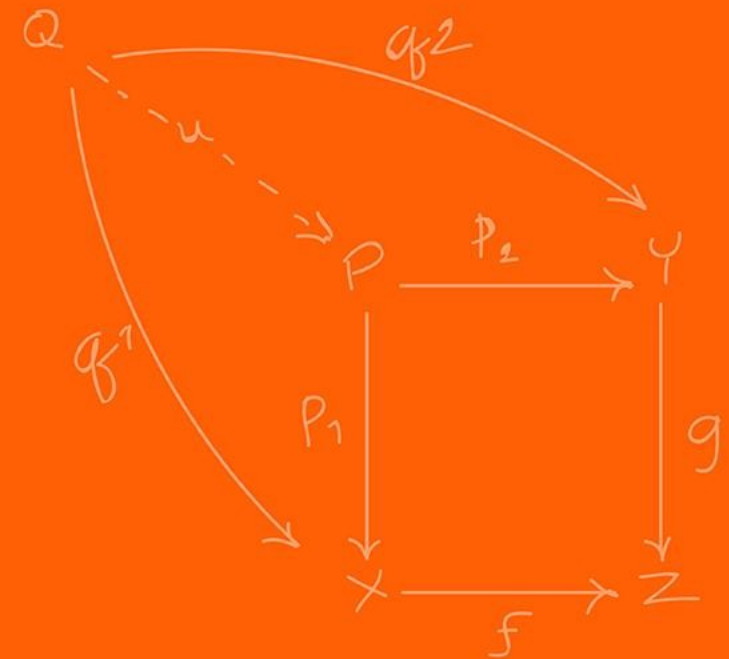
Pincushion



Mustache



Stereo Vision



How do we figure out how far an object is?

Keep both of your eyes open and then try to grab an object.

Now close one of your eyes and try picking it up. Does it feel different? Harder to judge the distance?

Now alternate closing the left and right eyes, do they both see the same thing? No? This is exactly what stereo vision exploits!

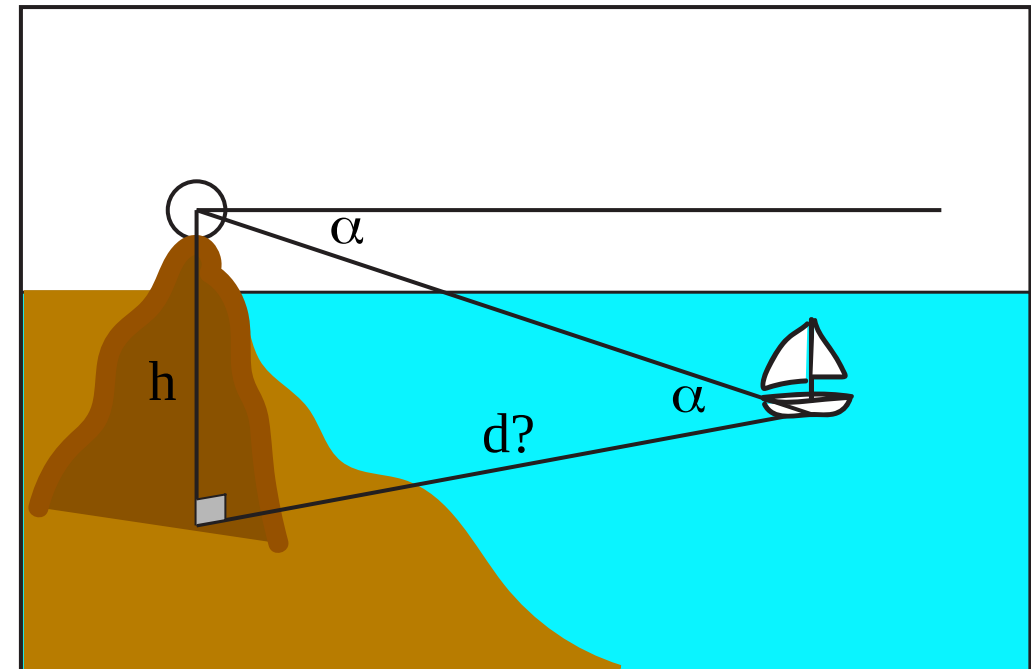
There are two typical methods to check for distance:

The first method measures the location of the boat relative to the horizon line. An observer is standing on top of an elevated point at a known height h .

The observer measures the angle α . You can derive the distance using simple trigonometric relation.

$$d = \frac{h}{\tan(\alpha)}$$

This method is not very accurate and might require high observation points, which might not be available, but it allows as to estimate with a single observation point.



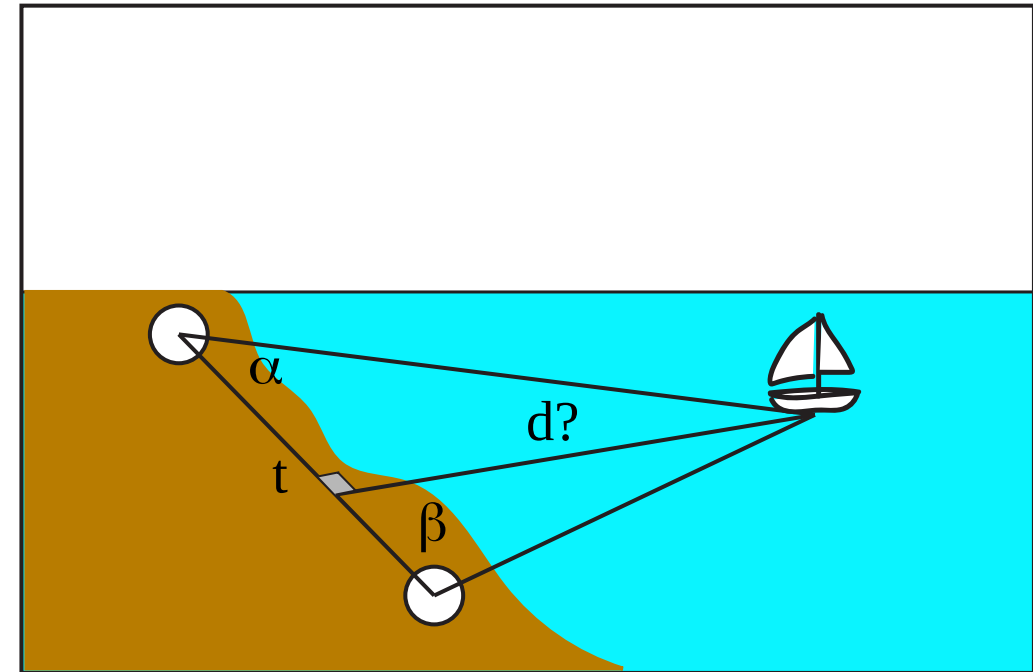
The second method requires two different observation points along the coast separated by a distance t .

At each point, we measure the angles, α and β , between the direction of the boat and the line connecting the two observation points.

Then we can apply the following relation:

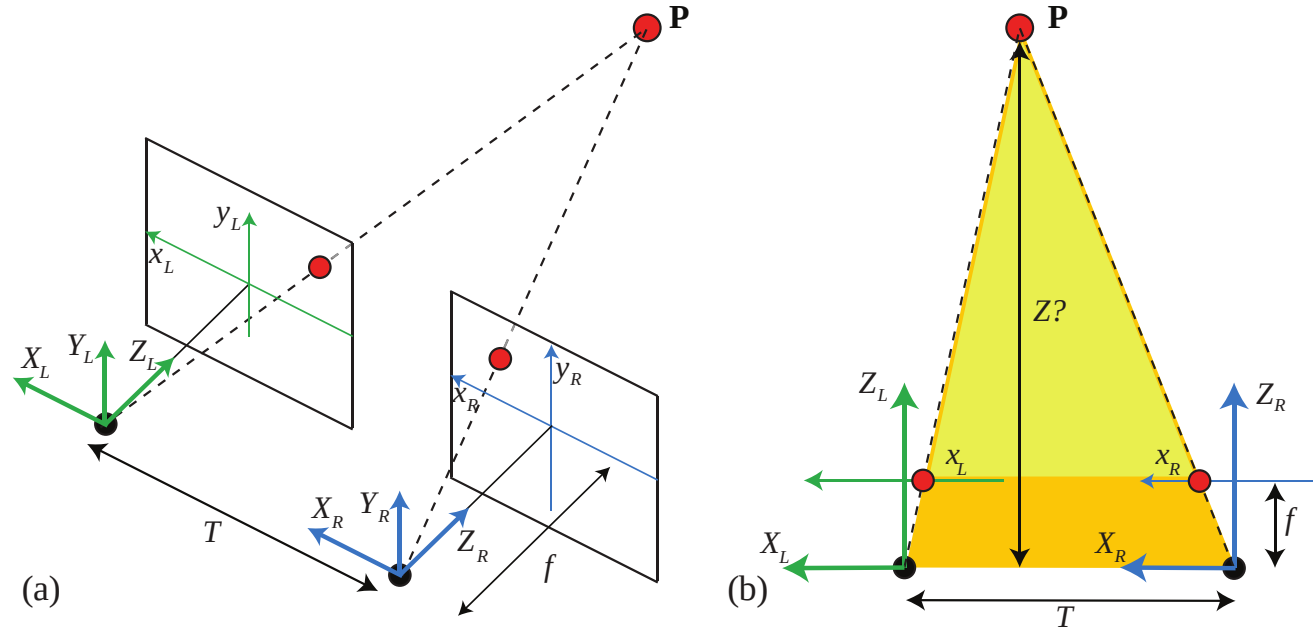
$$d = t \frac{\sin(\alpha) \sin(\beta)}{\sin \alpha + \beta}$$

This method is called **triangulation**. For it to work we need a large distance t so that the angles are significantly different from 90 degrees.



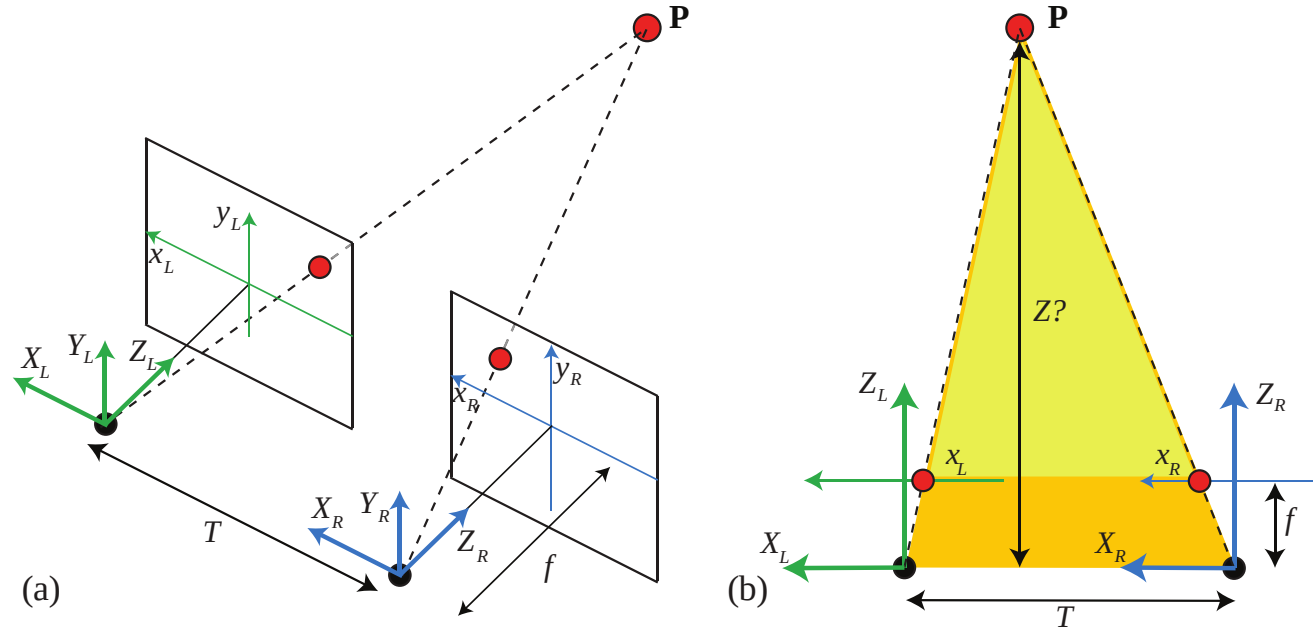
Stereo vision uses two cameras separated by a known baseline distance. By observing the same 3D point from two different views, we can triangulate its position.

When observing the world with two eyes we get two different views of the world. The relative displacement of features across the two images (**parallax effect**) is related to the distance between the observer and the observed scene.



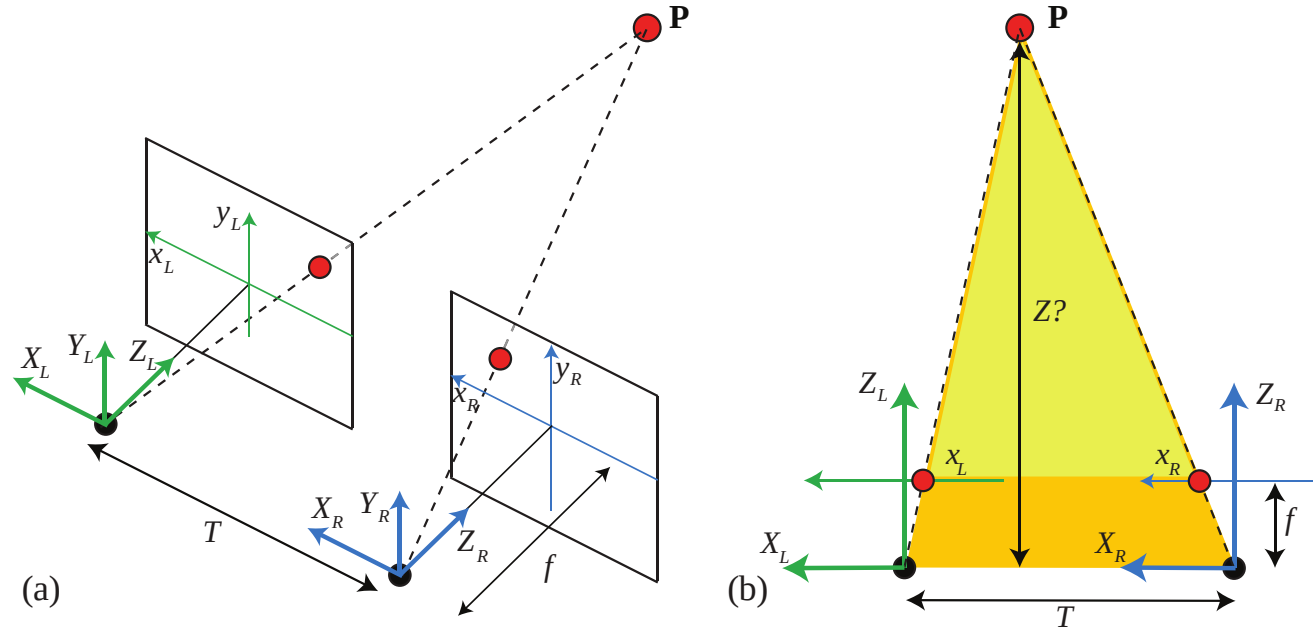
A simple way to triangulate the depth, Z , of a point visible in both images is through similar triangles. The two triangles shown in are similar (same angles), and so the ratios of their height to base must be the same. Thus we have:

$$\frac{T + x_r - x_L}{Z - f} = \frac{T}{Z}$$



Solving for Z we get that:

$$Z = \frac{fT}{x_L - x_R}$$



The difference in horizontal position of any given world point, P , as seen in the left and right camera images, $x_L - x_R$, is called the **disparity**, and is *inversely proportional* to the depth Z of the point P for this configuration of the two cameras—parallel orientation.

The task of inferring depth of any point in either image of a stereo pair thus reduces to measuring the disparity everywhere.

What is a problem in this situation in the real world?

One important assumption we have made is that we **know** the two corresponding projections of the point P ; however, in the real world the two images will be complex, and we will not know which points in one camera correspond to the same 3D point on the other camera.

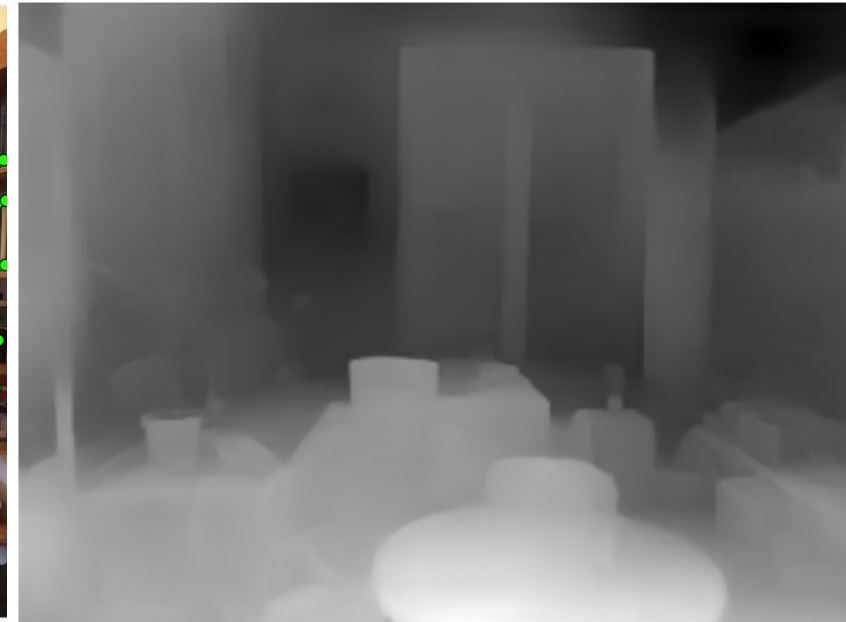
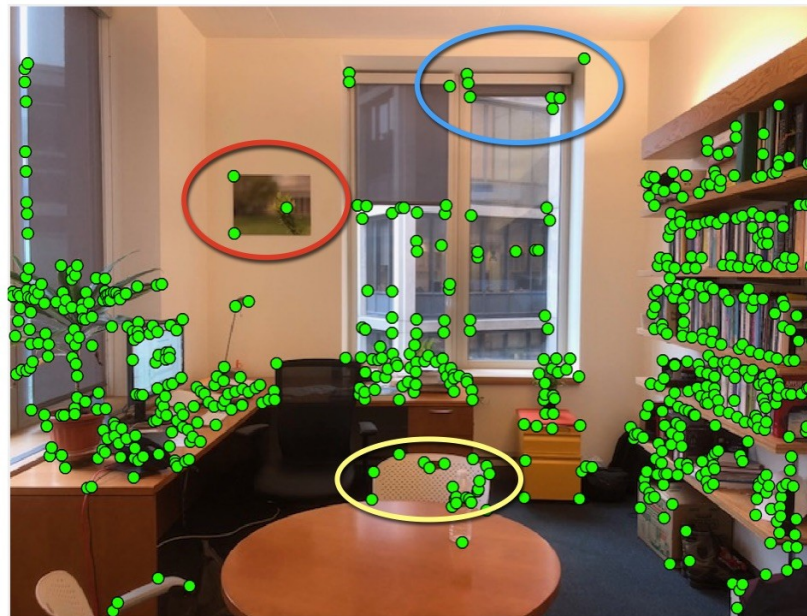
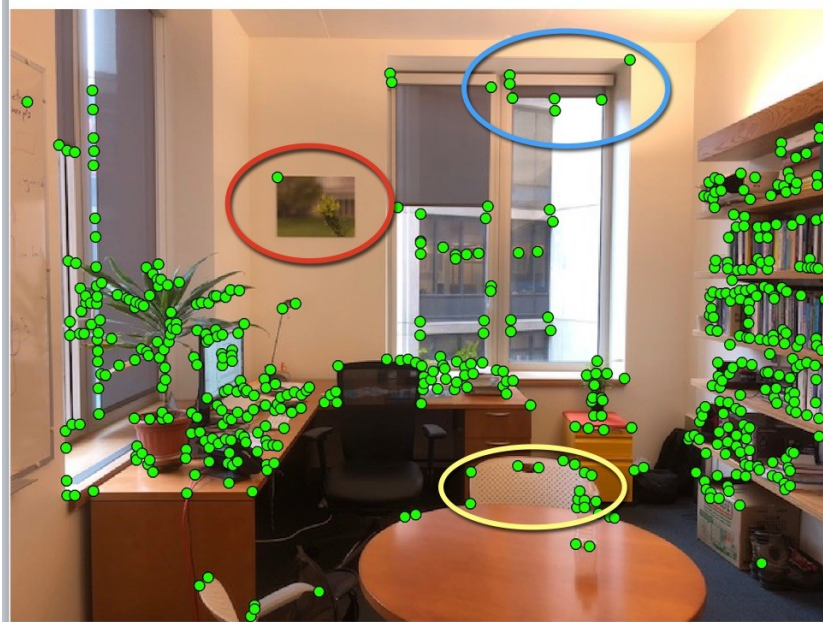
This is what the problem of **stereo matching** solves!

Stereo Matching

Stereo matching is the process of finding key points inside the images and finding the displacement between the two images.

Like your eyes, you don't necessarily notice individual pixels change but you can see key points (corners, objects). That have are different positions from one eye to another.

You then match the difference between the 2 images to match the points and retrieve P.



The task of finding disparity at each point is often broken into two parts:

1. **finding** features, and **matching** the features across images
2. **interpolating** between, or adjusting, the feature matches to obtain a reliable disparity estimate everywhere.

These two steps are often called matching and filtering, because the interpolation can be performed using filtering methods.

The Essential and Fundamental Matrices

The task of finding disparity at each point is often broken into two parts:

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IMAGE / GRAPHIC

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IMAGE / GRAPHIC

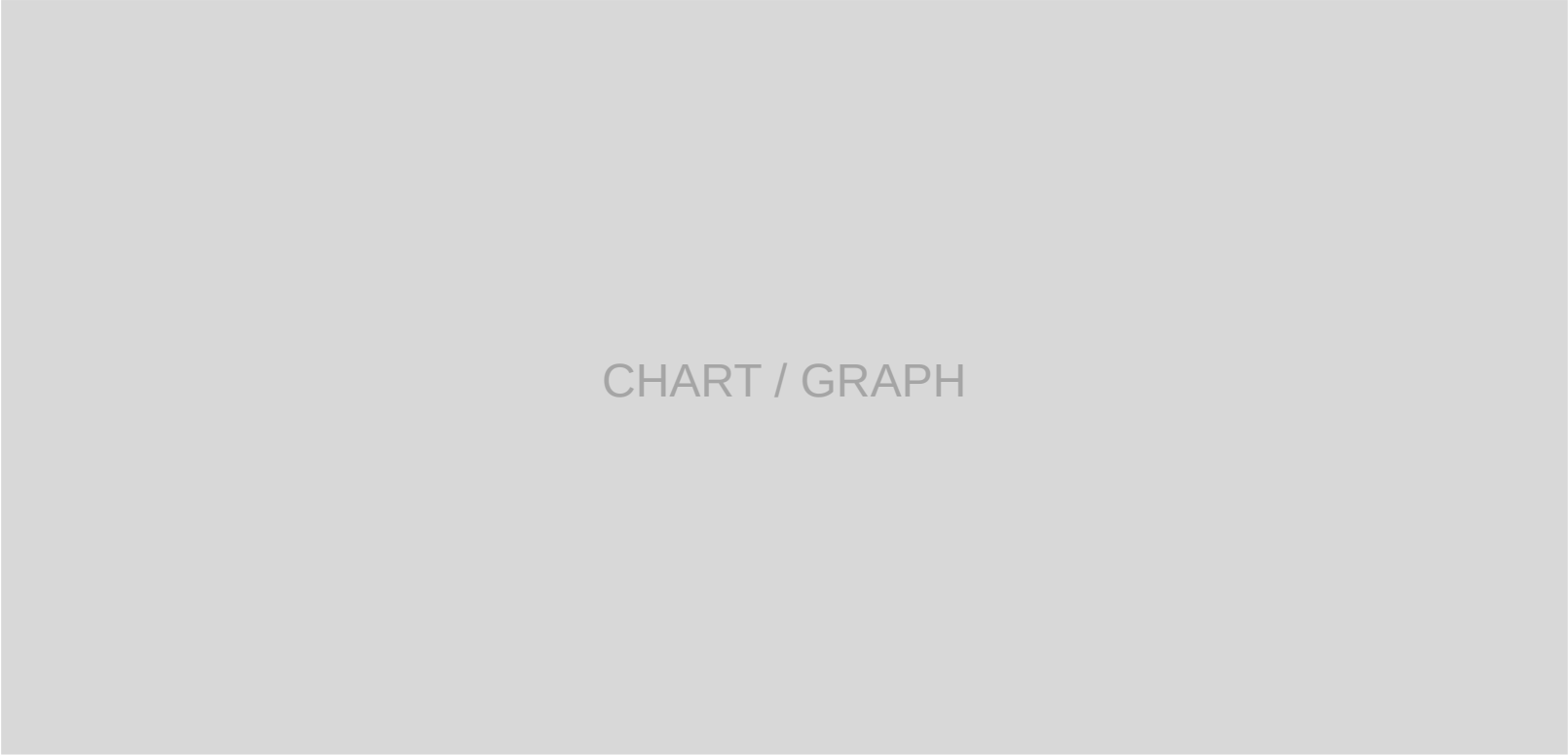
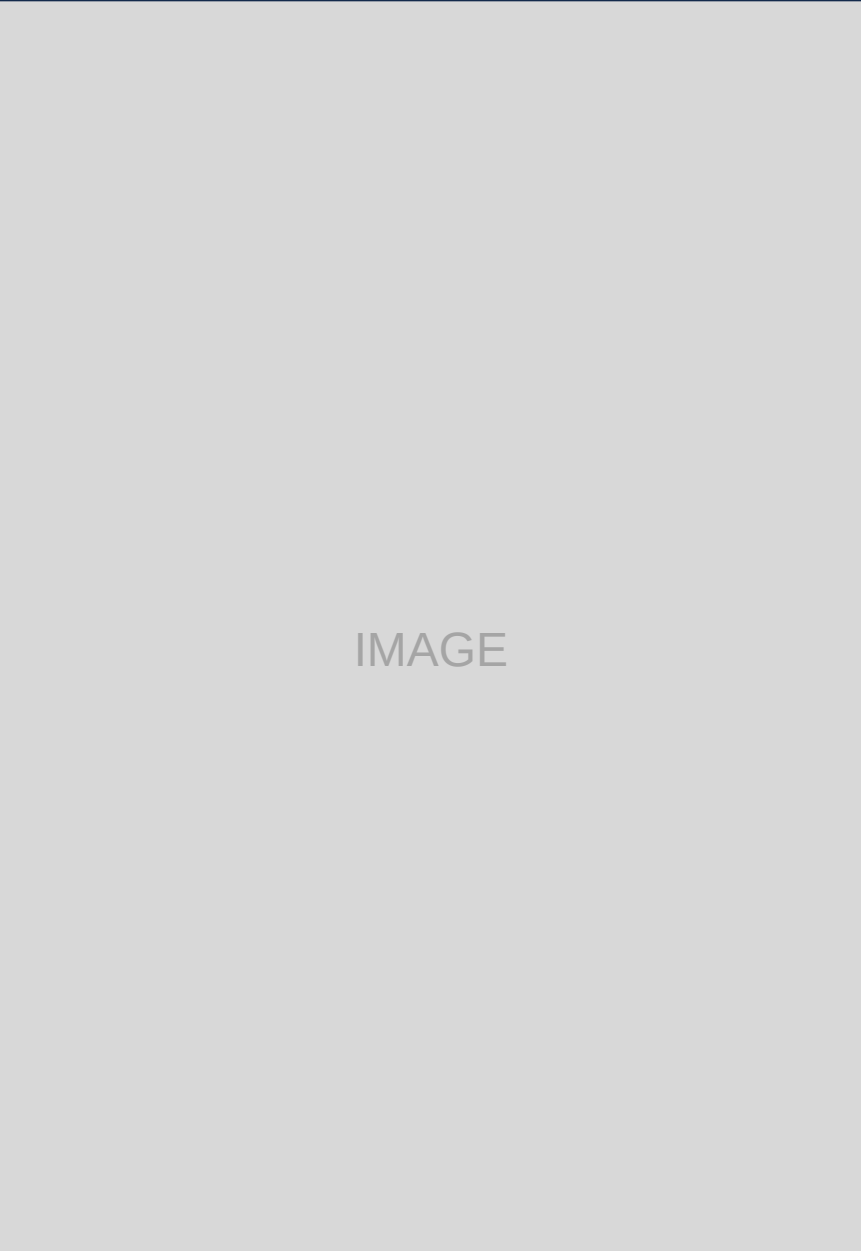


CHART / GRAPH

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IMAGE (Send to back so Block I is not covered.)

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(Keep this text simple and short on one or two lines.)



IMAGE

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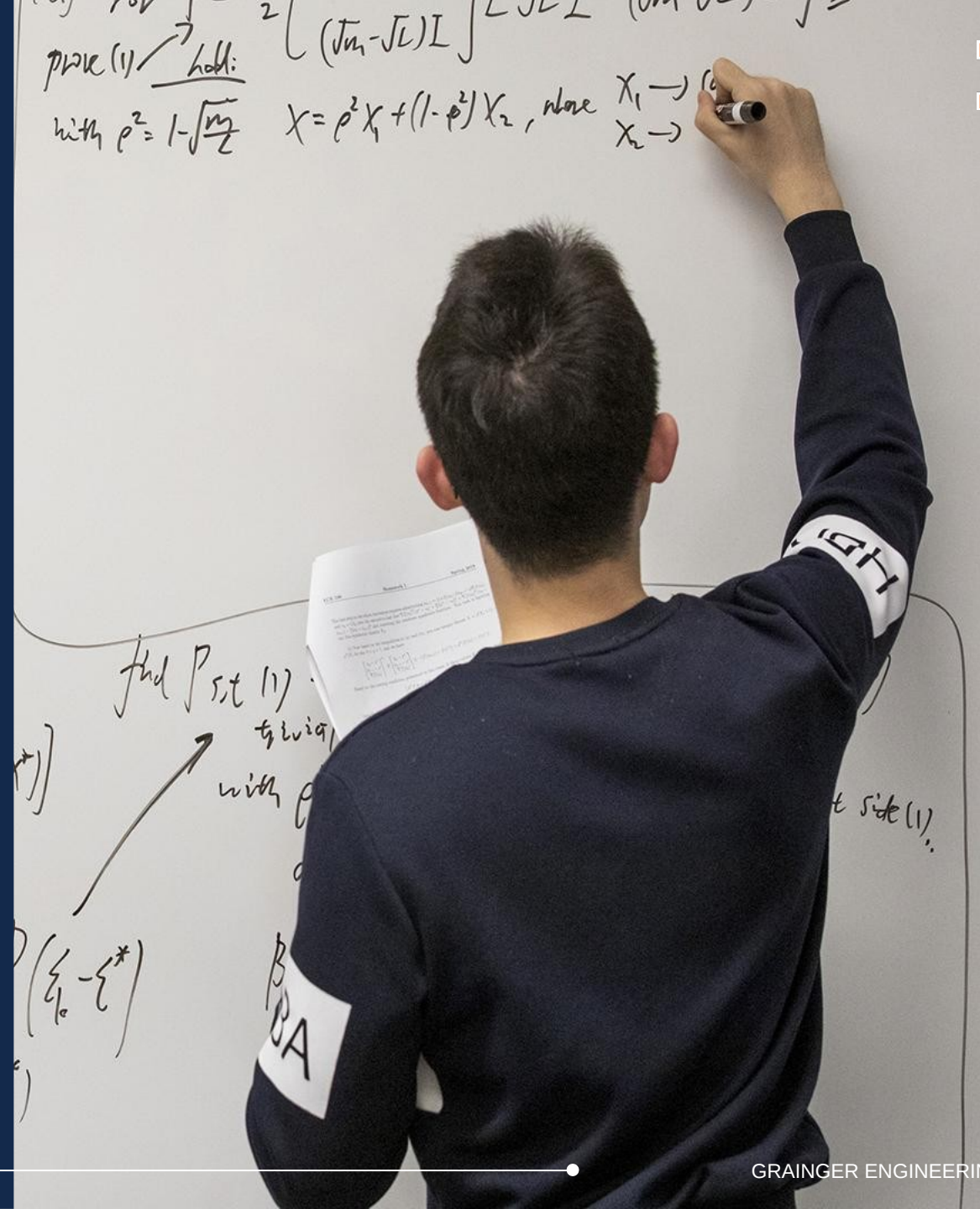
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IMAGE

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(Slide the blue box and the text left or right to fit over your background image. Use this text box for a call out or caption to the image.)



(Closing slide option. Suggestions for last slide)

Summary

Thank You

Questions

Contact Information

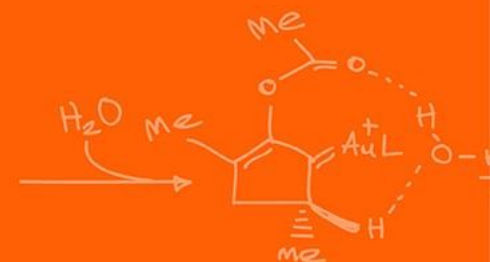
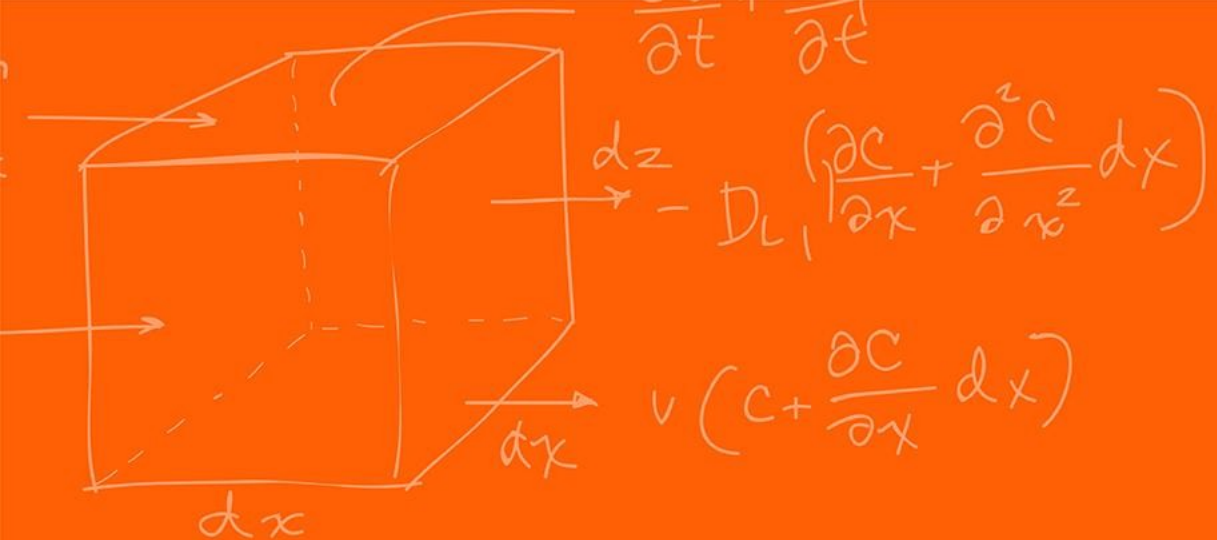
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Questions?



The Grainger College of Engineering

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